Deciding Membership and Realizability for Wheeler Transducers

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Wheeler transducers combine a total colexicographic order on reaching languages with a finite convex piecewise monotonicity condition on accumulated output. The number of monotone pieces is unspecified. We give a deterministic polynomial-time membership algorithm for finite trimmed deterministic subsequential transducers, allowing partial transitions and erasing outputs. This resolves the membership question for the piecewise definition of D'Agostino and Paradiso.

The structural criterion distinguishes two forms of recurrence. In a branching strongly connected component, a strict output turn can be replicated in disjoint input intervals, so finite piecewise monotonicity requires global monotonicity. A simple cycle instead has two sides approaching a central cut; each side is monotone exactly when two adjacent levels are monotone. These conditions reduce membership to finitely many comparisons of independent regular output ranges. A direct implementation takes $O(S^{13})$ bit operations for explicit encoding size S; this bound is conservative. Every output failure after input-order success has a common-loop pumped certificate of disjoint strict turns, with a total computable template bound.

As a consequence, equivalent Wheeler realizability is decidable and a realization can be constructed on positive instances. This uses classical prefix normalization and residual merging, an effective automatic convex quotient, and the membership algorithm. No complexity or minimality claim is made for the realization construction.

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1 The decision problems and the contribution

A subsequential transducer reads its input deterministically, emits a finite word on each transition, and may append a terminal word. D'Agostino and Paradiso define a Wheeler transducer by requiring both a total colexicographic order on its reaching languages and finite piecewise monotonicity of its accumulated output on each such language [5, 6]. Their current definition permits a different finite number of convex pieces at each state, with either orientation on each piece. Section 4 of both versions asks whether membership is decidable and whether an equivalent Wheeler realization can be found. The issue is effective control of an unbounded partition, rather than recognition for a prescribed number of pieces.

Our principal contribution is an exact criterion on strongly connected components (SCCs). After checking input order, global output monotonicity is decidable by propagating local violations. Branching cyclic SCCs must satisfy this global condition. A simple cycle permits a different phenomenon: two monotone sides can meet at an unrealized cut, even with infinitely many turns crossing that cut. We characterize its two sides by adjacent-level tests. Acyclic states then inherit finite partitions. All tests use finite automata for independent output ranges; no arbitrary intersection of rational relations is required. This supplies an algorithm, a polynomial bound, and effective obstructions for the original unrestricted membership problem.

For context, Wheeler *language* recognition from a DFA already has an $O(mn)$ algorithm [2]. That problem does not test accumulated output. Prefix normalization and minimization are classical [1], as are effective first-order queries and representatives in automatic structures [3, 4]. We use these as established tools for the realizability consequence. The statewise residual-prefix identity used there is already recorded in Remark 3 of [5]. The independent structural step here is the SCC criterion for finite convex output partitions.

1.1 Model and statements

Both finite alphabets Σ, Γ have specified total orders. For words, $u <_{\mathrm{co}} v$ means $\mathrm{rev}(u) <_{\mathrm{lex}} \mathrm{rev}(v)$, with a proper prefix smaller in lexicographic order. Thus a proper suffix is smaller in colexicographic order. A set is convex relative to an ordered domain if it contains every domain element between two of its elements. A function is *piecewise monotone* if its domain has a finite partition into nonempty convex sets on each of which it is weakly increasing or weakly decreasing.

Let $T = (Q, s, \Sigma, \Gamma, \delta, \omega, F, t)$ be finite and trimmed: every state is accessible and can reach a final state. The deterministic transition function δ is partial; ω has the same domain and takes values in Γ^*, including ε. There is no initial output. For a defined path, write $A(u) = \omega(s, u)$ and

$$I_q = \{u : \delta(s, u) = q\}, \qquad D = \bigcup_{q \in Q} I_q.$$

On accepted inputs the partial function is $f(u) = A(u)t(\delta(s, u))$. Trimming gives $D = \mathrm{Pref}(\mathrm{dom}\, f)$. A nonempty trimmed instance has a nonempty accepted domain. An encoding admitting an empty function can handle that degenerate case separately.

We use precisely the literal reaching-language definition in Definitions 4–5 of [5]: distinct states must satisfy either $I_p < I_q$ or $I_q < I_p$, where $X < Y$ abbreviates $\forall x \in X \forall y \in Y\ (x <_{\mathrm{co}} y)$; and $A|_{I_q}$ must be piecewise monotone for every q. No additional input-consistency or initial-indegree convention is imposed. Membership tests accumulated transition output A, not the terminal-output function f.

Theorem 1.1. *Membership in this class is decidable in $O(S^{13})$ bit operations, where $S \geq 2$ is the total explicit encoding size, counting states, transitions, ordered alphabet descriptions, and all output characters. Partial transitions, erasure, and arbitrary terminal words are allowed.*

Theorem 1.2. *Given such a transducer, it is decidable whether an equivalent finite Wheeler transducer over the same ordered alphabets exists. On a positive instance one can construct one, without introducing an initial output. No polynomial bound or minimality assertion is made for this construction.*

Section 4 proves the additional pumped-certificate assertion. All arguments below apply to infinite reaching languages. Finite controls described at the end supplement the proofs rather than establishing them.

2 Convex blocks and exact global monotonicity

We first check whether all reaching languages are pairwise separated. They are regular. Reverse their automata and use the regular-language comparison test below to decide each of the two possible universal strict orders. Reject a pair for which neither holds. Otherwise the induced state relation is a strict total order: transitivity follows directly from the universal definition and nonemptiness. Each I_q is then convex in D, since an intervening word in another state would contradict separation.

Lemma 2.1 (Recursive blocks). *Assume input order passes. The immediate ordered disjoint blocks of I_q are $\{\varepsilon\}$ if $q = s$, followed by all $I_p a$ with $\delta(p, a) = q$, sorted first by a and then by the order of p. Each is convex in I_q. More generally, if $\delta(p, z) = q$, then $I_p z$ is convex in I_q, and*

$$A(uz) = A(u)\omega(p, z) \quad (u \in I_p). \tag{1}$$

Common right concatenation preserves and reflects every colex comparison.

Proof. Every nonempty word has a unique last transition. Different last letters determine their colex order, and a common last letter cancels, leaving the separated predecessor languages. The empty word is least. This proves the immediate decomposition and convexity. Iterating it proves the assertion for z; the output identity follows from the path definition. Finally, reversal turns a common right suffix into a common left prefix, which cancels in lex order. This also holds for an empty suffix and proper-suffix comparisons. □

Let $O_p = \{A(u) : u \in I_p\}$. This is an effective regular language: start at s, accept at p, forget input labels, and read each transition's output word, subdividing word edges and retaining epsilon edges. A block $I_p z$ has output range $O_p\omega(p, z)$. Singleton blocks have singleton ranges. These ranges can be used independently when the input blocks are distinct and already ordered; no relation between their chosen words is required.

Lemma 2.2 (Regular range comparison). *For two epsilon-NFAs one can decide whether some accepted words x, y satisfy $x <_{\text{lex}} y$, $x >_{\text{lex}} y$, or $x = y$, in polynomial time. The same holds in colex order and for universal weak or strict order.*

Proof. Compute epsilon closures. Synchronize both automata on equal symbols until the first mismatch. A strict inequality is possible exactly when a reachable product state either permits two suitably ordered distinct next symbols with accepting continuations on both sides, or permits acceptance on the shorter side and a positive-length accepted continuation on the other. All these predicates are finite reachability tests. Equality uses simultaneous acceptance after a common word. Reversing automata gives colex comparison. Universal weak order excludes the opposite strict comparison; universal strict order excludes both the opposite comparison and equality. □

For $e \in \{+, -\}$, orientation + means weakly increasing and − weakly decreasing. Let S_e be the states with a violating pair from two distinct immediate blocks in their prescribed order. Lemma 2.2 decides these seeds. Let B_e contain all states reachable in the input graph from S_e, including seeds, and put $G_e(q) \iff q \notin B_e$.

Lemma 2.3. *$G_e(q)$ holds exactly when $A|_{I_q}$ has global orientation e.*

Proof. A seed violation propagates along every defined path by (1). Conversely, take a violating pair at q. If its words are in different immediate blocks, q is a seed. Otherwise they lie in one $I_p a$; neither can be the singleton epsilon block. Delete the common last letter. The pair remains strictly violating at p. Repetition terminates because the sum of lengths decreases, reaching a seed. The removed letters give a path from that seed to q. □

Write $M(q) = G_+(q) \vee G_-(q)$. We will also use two elementary order facts. First, a map between total orders is globally monotone in some orientation exactly when it has no strict peak or valley triple. Indeed, if a rising and a falling pair exist, sort their at most four arguments. Delete consecutive equal values. The successive strict signs include both orientations, so a sign change gives a strict turn. Second, infinitely many strictly separated closed strict-turn intervals preclude every finite convex monotone partition: each interval must cross a piece boundary, and a boundary can split at most one interval in such a family. Both increasing and decreasing families of intervals are covered.

3 The SCC criterion

SCCs refer to the ordinary input graph, counting distinct labeled edges. A cyclic SCC is *simple* if its internal graph is a directed simple cycle, including a single self-loop. A loop-free singleton is acyclic.

3.1 Branching recurrence

Lemma 3.1. *In a cyclic nonsimple SCC, finite piecewise monotonicity at every state is equivalent to $M(q)$ at every state.*

Proof. Sufficiency is immediate. For necessity, a nonsimple strongly connected component has a vertex r with two internal outgoing edges. Determinism makes their input letters distinct. Completing return paths gives nonempty loops u, v at r with distinct first letters, hence $uv \neq vu$. The infinite words $\mathrm{rev}(u)^\omega$ and $\mathrm{rev}(v)^\omega$ must differ: if equal to w, the two periods imply that both $\mathrm{rev}(u)\,\mathrm{rev}(v)$ and $\mathrm{rev}(v)\,\mathrm{rev}(u)$ are the same-length prefix of w. Effectively choose k large enough that $A_0 = u^k$ and $B_0 = v^k$ have a mismatch in reversal before either ends. They are suffix-incomparable.

If some q in the component fails $M(q)$, it has a strict turn. Appending a path from q to r transports that turn to words $\alpha_1 <_{\mathrm{co}} \alpha_2 <_{\mathrm{co}} \alpha_3$ reaching r. For $n \geq 0$ set

$$U_i(n) = \alpha_i A_0 B_0^n \quad (1 \leq i \leq 3). \tag{2}$$

The outputs retain the same strict turn by common suffix cancellation. The suffix cylinders of $A_0 B_0^n$ are strictly separated in a fixed direction. For consecutive indices, cancel the common reversed B_0^n; the comparison then encounters the fixed mismatch between $\mathrm{rev}(A_0)$ and $\mathrm{rev}(B_0)$, before any prefix α_i can matter. Thus either every interval precedes the next, or every interval follows it. The order fact above rules out a finite partition at r, proving necessity for the component property. □

3.2 Simple cycles and their two sides

Let C be a simple cyclic SCC. Choose $q \in C$, let c be the nonempty input word for one full traversal from q to itself, and put $b = \omega(q, c)$. Expand the unique internal predecessor block repeatedly for $|C|$ steps in Lemma 2.1. Retain that child and list all other children to its left or right. This yields

$$I_q = L \,\dot\cup\, I_q c \,\dot\cup\, R, \qquad L < I_q c < R. \tag{3}$$

Each side is a finite ordered list of singleton words arising from s, or blocks $I_p z$ with $p \notin C$ and z an explicit entering path of length at most $|C|$. There is only one internal predecessor at every step; this is why the construction stops after one traversal. Either side may be empty.

Put $L_n = Lc^n$ and $R_n = Rc^n$. Iteration of (3) gives

$$I_q = \left(\dot{\bigcup}_{n\geq 0} L_n\right) \dot{\cup} \left(\dot{\bigcup}_{n\geq 0} R_n\right). \tag{4}$$

There is no leftover finite word: membership in every $I_q c^n$ would require length at least $n|c|$ for all n. The L_n increase in the order, the R_n decrease, every left level precedes every right level, and each level and each whole side is convex in I_q.

Lemma 3.2 (Two-side test). *$A|_{I_q}$ is piecewise monotone if and only if each restriction*

$$A|_{L\cup Lc} \quad \textit{and} \quad A|_{Rc\cup R} \tag{5}$$

is globally weakly monotone, with directions chosen independently. If this holds, two pieces suffice at q.

Proof. If one restriction fails both orientations, it has a strict turn $v_1 <_{\mathrm{co}} v_2 <_{\mathrm{co}} v_3$. Appending c^{2n} places its three points on levels $2n$ and/or $2n+1$ of the same side. Consecutive pairs of levels are strictly separated, increasingly on the left and decreasingly on the right. The strict output turn persists after appending b^{2n}, including when $b = \varepsilon$. Infinitely many separated turn intervals contradict any finite partition.

Conversely, if the left restriction has orientation e, appending c^n proves the same orientation on $L_n \cup L_{n+1}$ for every n. If L is nonempty, all its levels are nonempty. For points on nonadjacent levels, insert one point in each intervening level and apply transitivity of the weak output order. Hence the whole left side has orientation e. An empty side needs no piece. The identical argument in decreasing level order applies on the right, using its independently chosen orientation. The two convex sides in (4) give the asserted partition. □

The test is effective. Expand (5) into its finite ordered lists, appending c to paths for the second copies. Orientation e holds on such a list exactly when every nonsingleton block $I_p z$ has $G_e(p)$, and every earlier/later pair of blocks has output ranges in weak order e. Lemma 2.1 checks the within-block condition exactly; the independent ranges and Lemma 2.2 check all cross-block pairs. Thus arbitrary regular prefix-output ranges are retained, without a depth bound or a bounded-expression assumption. All states of C inherit finite partitions from q: append a path from the chosen state to q, restrict the partition there, and use suffix cancellation. The preimages are convex.

The separation of the sides is essential. Consider the three-state machine with transitions $s \xrightarrow{0/\varepsilon} p$, $s \xrightarrow{1/\varepsilon} r$, $r \xrightarrow{0/\varepsilon} p$, $p \xrightarrow{0/0} p$, sole final state p, and empty terminal output. Here $I_p = 0^+ \cup 10^+$; $A(0^m) = A(10^m) = 0^{m-1}$. Its first part increases and its second decreases in colex. Infinitely many peak triples $0^{n+1}, 0^{n+2}, 10^{n+1}$, for $n \geq 0$, cross the same central cut, but the map has two pieces. A cyclic turn alone is therefore not a nonmembership certificate. Lemma 3.2 detects precisely the turns that can recur in disjoint levels.

3.3 Algorithm and proof of Theorem 1.1

Check input order, compute regular ranges and the two global flags, and compute SCCs. Require $M(q)$ at every state in each branching cyclic SCC. At one representative of each simple cyclic SCC require the two-side test. Reject on a failed requirement, and otherwise accept. Every rejection is sound by the preceding lemmas. Every passing cyclic SCC has finite partitions at all its states. The remaining components are loop-free singletons. In a topological order, their immediate blocks inherit the finite partitions of predecessor states, transported by their last transitions. Concatenating these partitions, with the initial singleton if present, gives a finite convex monotone partition. This proves sufficiency at every state. All constructions terminate; terminal words are unused. Empty input or output alphabets and the epsilon-only domain obey the same finite tests.

For completeness, a loose explicit resource bound follows. Write $n = |Q| \le S$, $m \le S$ for the edge count, and $M \le S$ for the maximum edge-output length. The range NFAs have $O(S)$ states and edges. There are at most $m + 1$ immediate blocks over all states. In one annular expansion per simple SCC, each external incoming edge occurs once, and the initial singleton occurs at most once overall. The doubled annular lists therefore have $O(S)$ blocks in total. Appended paths have length at most $2n$ and output length at most $2nM$. Every compared output NFA has size $O(S^2)$, and there are $O(S^2)$ comparisons; input-order tests use $O(n^2)$ pairs of $O(S)$-size input automata.

With K states and edges, straightforward epsilon elimination costs at most $O((|\Gamma| + 1)K^4)$ operations. Enumerating synchronized transitions, live unequal-symbol transitions, and prefix cases in a product comparison costs at most $O((|\Gamma| + 1)^2K^4)$. For $K = O(S^2)$ this is $O(S^{10})$ per comparison, hence $O(S^{12})$ operations overall. Allowing S bit operations per explicitly encoded symbol or index comparison gives $O(S^{13})$ bit operations. This deliberately conservative bound concerns membership only and proves Theorem 1.1.

4 Effective pumped failure certificates

Corollary 4.1. *For every input-ordered nonmember, some state has words $U_i(n) = x_i y_i^n z_i$, $1 \le i \le 3$, such that for every $n \ge 0$: all three reach that state, $U_1(n) <_{\mathrm{co}} U_2(n) <_{\mathrm{co}} U_3(n)$, their outputs form a strict peak for every n or a strict valley for every n, and either*

$$U_3(n) <_{\mathrm{co}} U_1(n+1) \text{ for all } n, \quad \text{or} \quad U_3(n+1) <_{\mathrm{co}} U_1(n) \text{ for all } n.$$

The three pumps can be chosen equal. There is a total computable bound $B(T)$ on the lengths of the nine template words needed for a certificate.

Proof. Every output rejection occurs in a branching or simple cyclic SCC. Equation (2) gives $x_i = \alpha_i A_0$, common $y_i = B_0$, and $z_i = \varepsilon$. The failed side in Lemma 3.2 gives $x_i = v_i$, common $y_i = c^2$, and $z_i = \varepsilon$. The proofs supply all universal inequalities, in either interval direction. Conversely any such certificate precludes a finite partition by separated turn intervals.

Run the membership algorithm first. On a rejection after input order, enumerate a finite strict-turn witness in the indicated state or regular two-level restriction; existence was proved, so this terminates. In the branching case the finite-power mismatch search also terminates. Output the maximum template length as $B(T)$. Set $B(T) = 0$ on acceptance, input-order failure, or a malformed or nontrimmed description. These cases are decidable, so B is total. No numerical complexity bound for B or the witness enumeration follows from the membership runtime. □

Validity of a fixed template is itself decidable, independently of that construction. For each fixed x_i, y_i, z_i, finite orbit decomposition of the state action of y_i gives a transient and a common period. On each residue, defined runs and their outputs have the form PQ^tR with fixed words, allowing $Q = \varepsilon$. A position has a given character precisely when it lies in a fixed prefix or suffix, or in the repeated block with its offset in a fixed residue modulo $|Q|$. All lengths are affine in t; when $Q = \varepsilon$ omit that block rather than taking a zero modulus. Strict lex comparison is a first differing position with equality before it, or a proper-prefix comparison. Thus input/output comparisons, with reversal for colex, are Presburger formulas using only constant coefficients and moduli. Finite transient checks and Presburger decidability verify every n, including consecutive-index separation. There is no multiplication of variables and no claim that a Presburger solver was executed. The polynomial membership algorithm does not use it.

5 Deciding equivalent realizability

We prove Theorem 1.2. The input here need not satisfy input order. Its partial function f and prefix domain D are fixed. The construction must merge redundant residuals *before* taking convex components: otherwise a redundant unary parity coloring of a constant function would create a false obstruction.

5.1 Normalization and canonical colors

Let $O_p(v)$ be the accepted continuation output from p, including its terminal word, and let $\lambda_p = \operatorname{lcp}\{O_p(v) : v \in \operatorname{dom} O_p\}$. Coaccessibility makes this range nonempty. Its output language is regular. Find one accepted output y and, among the finitely many prefixes of y, choose the longest z for which the entire range is contained in $z\Gamma^*$. Regular inclusion decides these tests and computes λ_p. For a transition $p \xrightarrow{a/w} r$, λ_p is a prefix of $w\lambda_r$, the LCP of all its accepted continuation outputs. It is also a prefix of $t(p)$ when p is final. Normalize to edge and terminal words

$$n(p,a) = \lambda_p^{-1}(w\lambda_r), \qquad n_t(p) = \lambda_p^{-1}t(p), \tag{6}$$

where inverse notation means deletion of the verified prefix. Telescoping gives normalized residual $g_p(v) = \lambda_p^{-1}O_p(v)$, with the same partial domain and with empty range LCP. This is classical prefix normalization [1].

Finite partition refinement computes equality of the g_p. Specifically, require equal final flags and normalized terminal words, equal defined letters and normalized edge words, and equivalent successors. These conditions suffice by induction on accepted continuation length. They are also necessary for equal functions: coaccessibility makes defined letters detectable from the partial domain, and the LCP of outputs on a-beginning continuations is exactly $n(p,a)$, since its target residual has empty LCP. Left cancellation then gives equal successor residuals. The greatest such finite relation is therefore exact. Write $[p]$ for these merged classes.

Create a distinct new initial state e with no incoming edges. Its terminal word is $f(\varepsilon)$ if defined. For each original edge $s \xrightarrow{a/w} r$, create $e \xrightarrow{a/(w\lambda_r)} [r]$; elsewhere use the normalized quotient. Delete unreachable classes, obtaining C. It is trimmed and computes exactly f without initial output. For every nonempty $u \in D$ its accumulated output is

$$A_C(u) = A_T(u)\lambda_{\delta(s,u)} = H(u) := \operatorname{lcp}\{f(uv) : uv \in \operatorname{dom} f\}. \tag{7}$$

At epsilon it remains ε, even if $H(\varepsilon) \neq \varepsilon$. Nonempty returns to the old initial state use its normalized class, not e. Hence initial cycles and nonempty initial LCPs are covered.

For nonempty u, the C-state is exactly its normalized continuation function $R_u(v) = H(u)^{-1}f(uv)$, on $u^{-1}\operatorname{dom} f$. Color each nonempty word by that state, and color epsilon by the unique color e. The colors are regular reaching languages. For any other trimmed realization W of f, with reaching set J_p and continuation LCP λ'_p, the statewise identity already used in [5, Remark 3] is

$$H(u) = A_W(u)\lambda'_p \qquad (u \in J_p). \tag{8}$$

It follows by factoring $A_W(u)$ from all accepted continuation outputs. Thus $J_p \setminus \{\varepsilon\}$ is monochromatic in the canonical coloring, regardless of output timing or the original machine's redundant states.

5.2 An effective convex quotient

Define $E(u,v)$ on D by equal canonical color and no word of a different color strictly between u and v in either direction. Its classes are exactly the maximal convex monochromatic components. Transitivity follows because the outer interval lies in the union of the intervals to a common point. Every convex monochromatic subset lies in a single component.

We detail the effective construction to distinguish synchronous regular relations from arbitrary rational relations. Encode words by reversal and use right-padded convolutions. Regular color predicates, equality, lex order, and shortlex order are synchronous regular. A lex comparator remembers the first differing symbol, treating an ended word as smaller; shortlex compares lengths first and then lex on equal lengths. Ended bits forbid a genuine symbol after padding. Boolean operations are relative to the regular universe of valid padded tuples. Existential projection guesses the removed track, including its end. Its witness can be longer than every free word: after projection, take the right quotient by an arbitrary all-padding suffix, equivalently marking a state final if it can reach acceptance along all-padding columns. This finite construction preserves padding invariance and imposes no length bound.

Apply these operations to

$$z \in D, \quad \operatorname{color}(z) \neq \operatorname{color}(u), \quad (u <_{\mathrm{co}} z <_{\mathrm{co}} v \ \vee \ v <_{\mathrm{co}} z <_{\mathrm{co}} u).$$

Project z, complement, and intersect with equal-color endpoints in D. This computes E. These are standard automatic-structure operations [3, 4]. Next compute the regular language

$$\operatorname{Rep}(u) \iff u \in D \ \wedge \ \neg\exists v \in D \ (E(u,v) \wedge \operatorname{rev}(v) <_{\mathrm{shortlex}} \operatorname{rev}(u)). \tag{9}$$

Shortlex is a well-order on a finite alphabet, so it selects exactly one representative per component, including components without a least colex word. Remove padding for the ordinary word encoding. A finite automaton decides finiteness of Rep: its reachable, coaccessible subgraph contains a genuine-letter cycle exactly when the language is infinite. If finite, enumerate its words in the useful acyclic graph. Thus the number of convex components is decidable without a guessed bound. Fixing a representative in E also gives an effective regular language for its component.

The relation E is right invariant. If uEv, their C-colors give the same defined letters and successor colors. For $u <_{\mathrm{co}} v$, any word $z \in D$ between ua and va ends in a, because that suffix cylinder is convex. Write $z = wa$. Prefix closure puts w in D, and suffix cancellation gives $u <_{\mathrm{co}} w <_{\mathrm{co}} v$. Hence w has the endpoint color and z the common successor color. Therefore $ua\,E\,va$. Equality and the reverse endpoint order give the other cases; partial definedness also agrees.

If Rep is finite, build a transducer C_c with one state per component, initial component $\{\varepsilon\}$, and the transition and terminal outputs of its C-color. At representative u, the target on a is the unique component whose representative r satisfies $E(ua, r)$. Right invariance makes the choice independent of u. Induction shows that every $w \in D$ reaches its own component with exactly $A_C(w)$, and the same accepted domain and terminal output as C. Every component is accessible and coaccessible. Thus C_c is finite, trimmed, equivalent to f, and its reaching languages are exactly the separated convex components.

5.3 Why no other realization can evade the tests

Suppose a finite Wheeler W computes f. Its reaching sets J_p are convex in D. Each $J_p \setminus \{\varepsilon\}$ is canonically monochromatic by (8), and removing epsilon preserves convexity because epsilon is least. These sets, with singleton epsilon, give a finite convex refinement of the canonical coloring. Therefore E has finitely many classes and Rep is finite.

Moreover, common right append of λ'_p preserves each orientation on each monotone piece of $A_W|_{J_p \setminus \{\varepsilon\}}$. Equation (8) transfers those pieces to H. Finitely many states and pieces thus give a finite convex monotone partition of H on $D \setminus \{\varepsilon\}$. Add singleton epsilon with output $A_C(\varepsilon) = \varepsilon$. By (7) this is a partition for A_C on D. Intersect it with each convex E-class. The resulting finitely many convex monotone pieces prove the output condition for C_c, whose input order already holds. Consequently *every* Wheeler realization forces this particular canonical convex split to pass membership.

The complete decision algorithm is now: construct C; compute E and Rep; reject if Rep is infinite; otherwise construct C_c and apply Theorem 1.1, returning C_c exactly on acceptance. All steps terminate. An infinite representative language rules out every Wheeler realization; a failed membership test also rules out every such realization by the preceding necessity argument. A positive answer supplies an equivalent Wheeler machine. This proves Theorem 1.2.

6 Verification, scope, and AI methodology

The proofs establish the claims for all promised finite inputs. Deterministic Python prototypes additionally implement regular-range comparison, seed propagation, SCC tests, normalization and residual merging, synchronous projection, regular representatives, finiteness, and the split construction. The membership controls covered all 544 accessible input-ordered binary partial machines through two states with outputs in $\{\varepsilon, 0, 1\}$, and 600 seeded machines with three to five states. All states were final. Direct sorted finite paths and a segmentation dynamic program checked the reported implications and piece bounds, through lengths seven and eight. Realizability controls included 13 boundary examples and 20 complete finite domains, checking 1,475 function values, 1,318 LCP identities, 8,171 relation comparisons, and 1,315 component-successor tests. Complete finite domains were compared against their entire direct convex decompositions; infinite controls used exact regular-language finiteness with supplementary literal replays through length seven.

Checks included erasure, partial terminal domains, the epsilon-only domain, redundant unary parity, nonempty initial LCP with an initial cycle, and finite canonical components whose output still fails membership. A separate model context reconstructed the arguments and ran additional finite checks. These activities are internal computational and mathematical checks, not external peer review or formal verification. No absence of a sampled counterexample is used as an infinite assertion. The accompanying local code and reports specify the finite coverage and retain failed approaches.

The membership exponent is not optimized, the certificate bound is only computable, and no practical indexing improvement, minimal realization, or realizability complexity bound is asserted. The results concern the stated piecewise definition with its literal reaching-language convention. The literature comparison is bounded by inspected primary sources; it is not a universal priority claim.

AI methodology. OpenAI Codex was used substantially throughout the project: it proposed and developed proof arguments, searched primary literature, generated and executed verification code, performed separate-context mathematical checks, and drafted and revised the exposition. This materially affected all technical sections, the computational controls, and the references. The checks do not constitute formal proofs or independent human peer review. No claim of an additional human verification is made here; the named author retains responsibility for the content and any future submission.

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